

# Healing

شِفَاءٌ

Doctor of Pharmacy 2016  
Calculus 102A  
Final Exam  
Dr.Issam Abu\_Irwaq



---

#نبض من الشفاء ارتوى

# \* Table of Integrals →

قواعد التكاملات

$$① \int k f(x) \cdot dx = k \int f(x) \cdot dx.$$

$$② \int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx.$$

→ التكامل يعطى على  
كلية الجمع - بطرح  
نقط...

$$③ \int k dx = kx + c. \quad \text{Ex: } \int dx = x + c$$

↓  
أحياناً مشكلة التكاملات  
أمرها للتوزيع على (\*، ÷)

$$④ \int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$⑤ \int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{(n+1)(a)} + c$$

→ مشتقة ما داخل القوس

$$⑥ \int \sin(ax+b) dx = \frac{-\cos(ax+b)}{(a)} + c$$

→ مشتقة الزاوية

$$⑦ \int \cos(ax+b) dx = \frac{\sin(ax+b)}{(a)} + c$$

→ مشتقة الزاوية

$$⑧ \int \sec^2(ax+b) dx = \frac{\tan(ax+b)}{a} + c$$

$$⑨ \int \csc^2(ax+b) dx = \frac{-\cot(ax+b)}{a} + c$$

$$⑩ \int \sec(ax+b) \tan(ax+b) dx = \frac{\sec(ax+b)}{a} + c$$

$$⑪ \int \csc(ax+b) \cot(ax+b) dx = \frac{-\csc(ax+b)}{a} + c$$

$$⑫ \int e^{ax+b} dx = \frac{e^{ax+b}}{a} + c$$

**MOHAMMAD  
AL\_ZOUBI**

$$⑬ \int \frac{x}{b} dx = \frac{bx}{2b} + c$$

$$⑭ \int \frac{1}{x} dx = \ln|x| + c$$

$$⑮ \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + c$$

Ex: Evaluate the following:

①  $\int 3^x dx.$

②  $\int (x^3 + \sqrt{x} + \frac{1}{x^5}) dx$

③

④  $\int (\tan^2(x)) \cdot dx$

⑤  $\int \tan(x) dx.$

⑥  $\int \sec(x) (\sec(x) + \tan(x)) \cdot dx$

⑦  $\int \frac{1}{1+\sin(x)} \cdot dx$

⑧  $\int \frac{1}{\sec(2x)} \cdot dx$

⑨  $\int e^{3 \ln x} dx.$

⑩  $\int e^{3 \ln x} dx$

⑪  $\int \sqrt{(4x+5)} dx$

⑫  $\int \sin^2(x) dx.$

---

①  $I = \int 3^x dx = \frac{3^x}{\ln 3} + c$

②  $I = \int (x^3 + (x)^{1/2} + x^{-5}) = \frac{x^4}{4} + \frac{x^{3/2}}{3/2} + \frac{x^{-4}}{-4} + c$

③

$$(4) \int (\sec^2 x - 1) dx \Rightarrow \tan(x) - x + C$$

$$(5) \int \frac{\sin(x)}{\cos(x)} dx \Rightarrow -\ln|\cos| + C$$

OR

$$\ln|\sec| + C$$

$$(6) \int \sec(x) (\sec(x) + \tan(x)) dx$$

$$= \int \sec^2(x) + \sec(x) \tan(x) dx$$

↓

$$\tan(x) + \sec(x) + C$$

$$(7) \int \frac{1}{1 + \sin(x)} dx \xrightarrow{\text{ملاحظة}} \int \frac{1}{1 + \sin(x)} * \frac{1 - \sin(x)}{1 - \sin(x)}$$

$$\int \frac{1 - \sin(x)}{\cos^2(x)} = \int \sec^2 - \sec \tan$$

$$\Rightarrow \int \sec^2(x) - \sec(x) \tan(x) dx$$

↓

$$\tan(x) - \sec(x) + C$$

-----

$$\int \frac{1}{2 \cos(x)}$$

هنا تم استخدام القريب بالمرافق

$$(8) \int \frac{1}{\sec(x)} dx \Rightarrow \int \cos(x) dx = \frac{\sin(2x)}{2}$$

$$(9) \int e^{3 \ln x} dx \Rightarrow \int e^{\ln x^3} \Rightarrow \int x^3 \Rightarrow \frac{x^4}{4} + C$$

ملاحظة: هنا استخدمنا تطبيق هذه خطوة عبان يكون  
عند  $1 = \ln$

## # The Integration by substitution

التكامل بالتعويض

Ex: Evaluate the following:

①  $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$

②  $\int \frac{\sin(\ln x)}{x} dx$

③  $\int \frac{1}{x(\ln x)^3} dx$

④  $\int \sec^3(\tan x) dx$

⑤  $\int (x+2) \sqrt{x^3+4x+11} dx$

⑥  $\int x \sqrt{x+1} dx$

السؤال: متى تستخدم التكامل بالتعويض..

1. إذا كانت دوايا، لاقترانات الدائرية غير خطية.

2. إذا كانت الأسس، في لاقترانات الأسية غير خطية.

3. إذا اخذت المسئلة على اللوغاريتمات نفرض  $\ln x = y$

4. إذا طالت مشتقة أحد الاقترانيين موجودة.  
حل سؤال رقم (5).

5. أسئلة الجذور غالباً نفرض.

MOHAMMAD  
AL\_ZOUBI

## Solution

① Set  $u = \sqrt{x} \Rightarrow \frac{du}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow dx = 2\sqrt{x} du$

$$\therefore \int \frac{e^u \cdot 2\sqrt{x} du}{\sqrt{x}} = 2 \int e^u \cdot du = 2e^u + C$$
$$= 2e^{\sqrt{x}} + C$$

② Set:  $u = \ln x \Rightarrow \frac{du}{dx} = \frac{1}{x} \Rightarrow dx = x du$

$$\therefore \int \frac{\sin(u) \cdot x du}{x} = -\cos(u) + C$$
$$= -\cos(\ln x) + C.$$

③ Set:  $u = \ln x \Rightarrow \frac{du}{dx} = \frac{1}{x} \Rightarrow dx = x du$

$$\Rightarrow \int \frac{x \cdot du}{x(u)^2} = \int u^{-2} \cdot du = \frac{u^{-1}}{-1} + C$$
$$= \frac{-(\ln x)^{-1}}{-1} + C$$

MOHAMMAD  
AL-ZOUBI

④ \* Set:  $u = \sec(x) \Rightarrow \frac{du}{dx} = \sec(x) \tan(x)$   
 $dx = \frac{du}{\sec(x) \tan(x)}$

$$\int \frac{\sec(x) \cdot \tan(x) \cdot du}{\sec(x) \tan(x)} = \int u^2 \cdot du$$

$$\frac{u^3}{3} + C = \frac{(\sec)^3}{3} + C$$

⑤ Set:  $u = x^2 + 4x + 1 \Rightarrow \frac{du}{dx} = 2x + 4 = 2(x + 2)$

$$\int \frac{(\cancel{x+2}) \cdot du}{2(\cancel{x+2})} = \int \frac{\sqrt{u}}{2} \cdot du = \frac{1}{2} \cdot \frac{u^{3/2}}{3/2} + C = \frac{\sqrt{x^2 + 4x + 1}^3}{3} + C$$

⑥ Set:  $u = \sqrt{x+1} \Rightarrow \frac{du}{dx} = \frac{1}{2\sqrt{x+1}}$   $u^2 = x+1 \rightarrow x = u^2 - 1$   
 $2u du = dx$

$$= \int (u^2 - 1) \cdot u \cdot du \Rightarrow \int (u^3 - u) \cdot u du = \frac{u^4}{4} - \frac{u^2}{2} + C \dots$$

[6]

# \* Integration of rational function by partial fraction

الأمثلة البسيطة

$$f(x) = \frac{p(x)}{q(x)}$$

where  $p(x)$  and  $q(x)$  are polynomials provided  $q(x) \neq 0$ .

for examples:  $f(x) = \frac{1}{x}$  ,  $f(x) = \frac{2x+1}{x^3-x}$

$$f(x) = \frac{x^2}{x+1}$$

Case I:

①  $\deg(p(x)) < \deg(q(x))$ .

②  $q(x)$  can be written as a product of distinct linear factors.

For examples:  $\frac{1}{x^2-4} = \frac{1}{(x-2)(x+2)}$

$$\frac{2x+1}{x^3-x} = \frac{2x+1}{x(x-1)(x+1)}$$

Ex: write the decomposition of partial functions for  $\frac{2x+1}{x^3-x}$ .

Step (1):  $\frac{2x+1}{x^3-x} = \frac{\text{constant}}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$

Step (2):  $\frac{2x+1}{x(x-1)(x+1)} = \frac{A(x-1)(x+1) + B(x)(x+1) + C(x)(x-1)}{x(x-1)(x+1)}$

Step (3):  $2x+1 = A(x-1)(x+1) + B(x)(x+1) + C(x)(x-1)$

\* هنا يتم تعريف أمتار المقام لكي أخرج قيم (C, B, A) و هذا أفضل طريقة

partial function :-

$$\frac{1}{x} + \frac{1}{x-1} = \frac{x-1+x}{(x)(x-1)} = \frac{2x-1}{x^2-x}$$

\* unlike fraction to like fraction  $\rightarrow$  تحويل كسور مختلفة المقامات إلى كسور متماثلة المقامات

\*  $\leftarrow$  partial fraction كسر جزئية

$$\Rightarrow 2x+1 = A(x-1)(x+1) + B(x)(x+1) + C(x)(x-1)$$

$$\underline{\text{Set } x=0} \Rightarrow 2(0)+1 = A(0-1)(0+1) \Rightarrow 1 = -A \Rightarrow \boxed{A = -1}$$

$$\underline{\text{Set } x=1} \Rightarrow 2(1)+1 = B(1)(1+1) \Rightarrow \boxed{B = \frac{3}{2}}$$

$$\underline{\text{Set } x=-1} \Rightarrow 2(-1)+1 = C(-1)(-1-1) \Rightarrow \boxed{C = -\frac{1}{2}}$$

Ex: Evaluate  $\int \frac{2x+1}{x^3-x}$

$$\underline{\text{Sol:}} \int \frac{2x+1}{x^3-x} = \int \frac{2x+1}{(x)(x-1)(x+1)} = \int \frac{-1}{x} + \frac{3/2}{x-1} + \frac{-1/2}{x+1}$$

$$= -\ln|x| + \frac{3}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C$$

OR

$$\ln \left| \frac{1}{x} \right| + \ln \left| \sqrt{(x-1)^3} \right| + \ln \left| \frac{1}{\sqrt{x+1}} \right| + C$$

OR

$$\ln \left| \frac{\sqrt{(x-1)^3}}{(x)(\sqrt{x+1})} \right| + C \dots$$



practice: For  $f(x) = \frac{1}{x^2-4}$   
a) write the decomposition of partial fraction for  $f(x)$ .

b) Find  $\int \frac{1}{x^2-4} \cdot dx$

Solution:  $f(x) = \frac{1}{(x-2)(x+2)} = \frac{A}{(x-2)} + \frac{B}{(x+2)}$

$$1 = A(x+2) + B(x-2)$$

set  $x=2$ :  $1 = 4A \Rightarrow A = \frac{1}{4}$

set  $x=-2$ :  $1 = -4B \Rightarrow B = -\frac{1}{4}$

$$b) \Rightarrow \int \frac{1}{(x-2)(x+2)} = \int \frac{1/4}{(x-2)} + \frac{-1/4}{(x+2)}$$

$$\Rightarrow \frac{1}{4} \ln|x-2| - \frac{1}{4} \ln|x+2| + C$$

$$\ln \left| \frac{\sqrt[4]{x-2}}{\sqrt[4]{x+2}} \right| + C$$

$$\ln \left| \frac{\sqrt[4]{x-2}}{\sqrt[4]{x+2}} \right| + C$$

$$\ln \left| \sqrt[4]{\frac{(x-2)}{(x+2)}} \right| + C$$

Question (6 point) (find):

Find  $\int \frac{3x+1}{x^2+x} dx$

1st Approach:

$$\frac{3x+1}{(x)(x+1)} = \frac{A}{x} + \frac{B}{(x+1)}$$

$$\frac{3x+1}{(x)(x+1)} = \frac{A(x+1) + B(x)}{x(x+1)}$$

$$\Rightarrow 3x+1 = A(x+1) + B(x)$$

$$\text{Set } x = -1 \Rightarrow 3(-1)+1 = B(-1) \Rightarrow \boxed{B = 2}$$

$$\text{Set } x = 0 \Rightarrow 3(0)+1 = A(0+1) \Rightarrow \boxed{A = 1}$$

$$\therefore \frac{3x+1}{(x)(x+1)} = \frac{1}{x} + \frac{2}{x+1}$$

$$\int \frac{3x+1}{x^2+x} dx = \int \frac{1}{x} + \frac{2}{x+1} = \ln|x| + 2\ln|x+1| + C$$

But:  $\int \frac{3x+1}{x^2+x} dx \Rightarrow \ln|x| + 2\ln|x+1|$

$$\begin{aligned} & \left| \begin{aligned} & \rightarrow \ln 3 + 2\ln 4 - \ln 1 \\ & = \ln 3 + 2\ln 2 \\ & = \ln 3 + 2\ln 4 - 0 \\ & \ln 3 + 2\ln 4 - \ln 1 \\ & \ln 3 + \ln 4 = \ln 12 \end{aligned} \right. \end{aligned}$$

2nd Approach:  $\int \frac{3x+1}{x^2+x} dx = \int \frac{2x+1+x}{x^2+x} = \int \frac{2x+1}{x^2+x} + \int \frac{x}{x^2+x}$

$$= \ln|x^2+x| + \ln|x+1| + C$$

$$\Rightarrow \ln|x^2+x| + \ln|x+1|$$

**MOHAMMAD  
AL-ZOUBI**

$\ln 12$

← طريقة سريعة لا يجد A, B ... دون الحاجة إلى طريقة انصبة

Ex:  $\int \frac{4}{(3x+4)(5x+8)} \cdot dx = 4 \int \frac{1}{(3x+4)(5x+8)} \cdot dx$

$\Rightarrow 4 \left[ \int \frac{\overset{+ \text{ناتج } 3}{\underset{\text{ناتج } 5}{3 \cdot 8 - 4 \cdot 5}}}{(\overset{3}{3}x + \overset{4}{4})} + \frac{\overset{5 \cdot 4 - 8 \cdot 3}{5x + 8}}{(\overset{5}{5}x + \overset{8}{8})} \right]$

$\Rightarrow 4 \left[ \int \left( \frac{3/4}{(3x+4)} + \frac{-5/4}{5x+8} \right) \cdot dx \right]$

$\Rightarrow 4 \left[ \frac{1}{4} \int \frac{3}{3x+4} + \frac{1}{4} \int \frac{5}{5x+8} \right] \cdot dx$

$\ln|3x+4| - \ln|5x+8| + C$

Case II: If  $\deg(p(x)) \geq \deg(q(x))$ .

Use long-dividing.

Ex:  $\int \frac{x^3}{x^2-1} \cdot dx$

$\int \left( \text{Result} + \frac{\text{Remainder}}{\text{divisor}} \right) \cdot dx$

$$\begin{array}{r} x \\ x^2-1 \overline{) x^3} \\ \underline{\pm x^3} \quad x \\ \hline x \end{array}$$

$\int \left( x + \frac{x}{x^2-1} \right) \cdot dx$

تصديق - الثاني  
يوم الخميس ١٤ / ٥ / ٢٠١٦  
موقعه

التوقيع :  
(٩٥) - ٣٩ (٧٢) SFD  
٨٠



$$\int \frac{x}{x^2 + 4x} dx$$

$$\text{Set : } a = 1 + \ln(x)$$

$$= \frac{x^2}{x^2} = 1$$



$$\int \frac{x}{u} \cdot du = \int \frac{1}{u} \cdot du$$

$$\ln|u| + C = \ln|1 + \ln u| + C - 7 + 10 \ln 7 + 1$$

# \* Integration by parts

بشكل بالجزء...

$$\int u \, dv = uv - \int v \, du.$$

Ex: Evaluate:

①  $\int x \ln(x) \, dx$       ②  $\int x^3 \ln(x) \, dx$       ③  $\int_1^e \ln(x) \, dx$

④  $\int x e^{-x} \, dx$       ⑤  $\int \ln x^x \, dx$       ⑥  $\int x \sin(x) \, dx$

⑦  $\int x^3 e^{2x} \, dx$       ⑧  $\int (x^3 - 5x + 7) \sin(x) \, dx$

⑨  $\int x^3 e^{x^2} \, dx$   
تقسيم + تجزئة

Solution: ①

$\begin{array}{l} u \\ \ln(x) \\ \frac{1}{x} \end{array} \quad \begin{array}{l} dv \\ x \\ \frac{x^2}{2} \end{array} \quad \left| \begin{array}{l} \Rightarrow \frac{x^2}{2} \ln(x) - \int \frac{x}{2} \\ \frac{x^2}{2} \ln(x) - \frac{x^2}{4} + C \end{array} \right.$

③

$\begin{array}{l} u \\ \ln(x) \\ \frac{1}{x} \end{array} \quad \begin{array}{l} dv \\ 1 \\ x \end{array} \quad \left| \begin{array}{l} \Rightarrow (x) \ln(x) - \int_1^e = (x) \ln(x) - x \Big|_1^e \\ (e \ln(e) - e) - (1 \ln(1) - 1) \\ = 1 \end{array} \right.$

④

⊕

$\begin{array}{l} u \\ \frac{1}{x} \\ 1 \end{array} \quad \begin{array}{l} dv \\ x \\ \frac{e^{-x}}{(-1)^{x-1}} \end{array}$

⊖

⊕

$\Rightarrow (x)(-e^{-x}) - (1)(\frac{e^{-x}}{-1}) + C$

هذه الخطوات اذا كانا

الجزء اربعة

MOHAMMAD  
AL ZOUBI

| * ⑥ | <u>u</u> | <u>du</u> |
|-----|----------|-----------|
| ⊕   | x        | sin(x)    |
| ⊖   | 1        | -cos(x)   |
| ⊖   | 0        | -sin(x)   |

$$\Rightarrow (x)(-\cos(x)) - (1)(-\sin(x)) + C$$

$$\Rightarrow -x \cos(x) + \sin(x) + C$$

**MOHAMMAD  
AL\_ZOUBI**

| ⑧ | <u>u</u>                  | <u>du</u> |
|---|---------------------------|-----------|
| ⊕ | (x <sup>3</sup> - 5x + 7) | sin(x)    |
| ⊖ | (3x <sup>2</sup> - 5)     | -cos(x)   |
| ⊕ | (6x)                      | -sin(x)   |
| ⊖ | (6)                       | cos(x)    |
| ⊕ | (0)                       | sin(x)    |

$$\Rightarrow (x^3 - 5x + 7)(-\cos(x)) - (3x^2 - 5)(-\sin(x)) + (6x \cos(x)) - 6 \sin(x) + C$$

$$\Rightarrow -(x^3 - 5x + 7)(\cos(x)) + (3x^2 - 5)(\sin(x)) + (6x \cos(x)) - 6 \sin(x) + C$$

properties of definite Integrals

خواص التكامل المحدد

$$① \int_a^b c f(x) \cdot dx = c \int_a^b f(x) \cdot dx$$

$$② \int_a^b (f(x) \pm g(x)) \cdot dx = \int_a^b f(x) \cdot dx \pm \int_a^b g(x) \cdot dx$$

$$③ \int_a^b c \cdot dx = c(b-a) \quad ④ \int_a^a f(x) \cdot dx = 0$$

$$⑤ \int_a^b f(x) \cdot dx = - \int_b^a f(x) \cdot dx \quad ⑥ \int_a^b f(x) \cdot dx = \int_a^c f(x) \cdot dx + \int_c^b f(x) \cdot dx$$

EX(1): Let  $\int_0^1 f(x) \cdot dx = \frac{4}{3}$  and  $\int_1^3 f(x) \cdot dx = -5$   
Find  $\int_0^3 f(x) \cdot dx$

EX(2): If  $\int_0^3 f(x) \cdot dx = 4$  and  $\int_1^4 2f(x) \cdot dx = 6$   
Find  $\int_0^4 f(x) \cdot dx$

MOHAMMAD  
AL\_ZOUBI

EX(3): Evaluate  $\int_{-1}^2 |x| \cdot dx$

EX(4): Let  $f(x) = \begin{cases} x^2 & \text{if } x \geq 2 \\ x-1 & \text{if } x < 2 \end{cases}$  Find  $\int_0^3 f(x) \cdot dx$ .

EX(5): Find the value of  $k$  so that  
 $\int_1^k k \cdot dx = 0$ .

EX(6): If  $f(x) = 2$ ,  $f(x) = 6$ , Find  $\int_0^2 x \cdot f(x) \cdot dx$ .

when  $x < 0 \Rightarrow y = 0$   
when  $x = 1 \Rightarrow y = 3$

solution (ex):

①  $\int_0^3 f(3x) \cdot dx = \frac{4}{3}$ , But: set:  $u = 3x \Rightarrow dx = \frac{du}{3}$

$\Rightarrow \int_0^3 \frac{f(u)}{3} \cdot du = \frac{4}{3} \Rightarrow \int_0^3 f(u) \cdot du = 4$

$\therefore \int_0^3 f(x) = \int_0^3 f(x) \cdot dx + \int_3^4 f(x) \cdot dx = \underline{4+5=9}$ .

②  $\int_0^4 f(x) \cdot dx = \int_0^3 f(x) + \int_3^4 f(x) \cdot dx$

$4 + 3 = 7$

③

$$\int_{-1}^2 |x| \cdot dx = \int_{-1}^0 -x \cdot dx + \int_0^2 x \cdot dx = \int_0^1 x \cdot dx + \int_0^2 x$$

$$\Rightarrow \frac{x^2}{2} \Big|_0^1 + \frac{x^2}{2} \Big|_0^2 \Rightarrow \frac{1}{2}((-1)^2 - 0^2) + \frac{1}{2}(4 - 0)$$

$$= \frac{1}{2} + 2 = \frac{5}{2}$$

④

$$\int_0^3 f(x) \cdot dx = \int_0^2 f(x) \cdot dx + \int_2^3 f(x)$$

$$= \int_0^2 (x-1) \cdot dx + \int_2^3 x^2 \cdot dx$$

$$= \left( \frac{x^2}{2} - x \right) \Big|_0^2 + \frac{x^3}{3} \Big|_2^3 \Rightarrow (0 - 0) + (9 - \frac{8}{3})$$

$$= 9 - \frac{8}{3} = \frac{27-8}{3} = (19/3)$$

⑤

$$\int_1^K x \, dx = 0$$

$K=1$  ,  $K(K-1)=0$   
 $K=0, K=1$

⑥

$$\frac{u}{x} \quad \frac{du}{f(x)}$$

$$1 \quad \leftarrow \quad f(x)$$

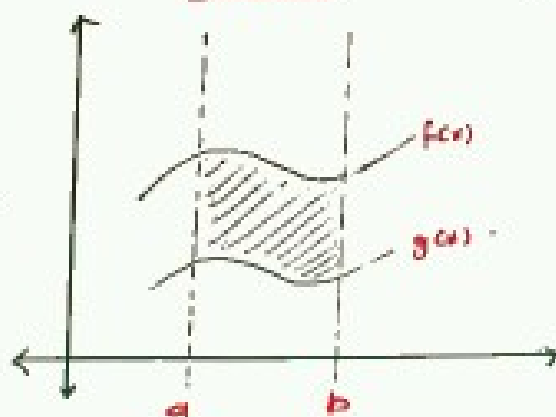
$$x f(x) \Big|_0^2 - \int_0^2 f(x) \cdot x \, dx = 2 f(2) - \int$$



MOHAMMAD  
AL\_ZOUBI

Area:

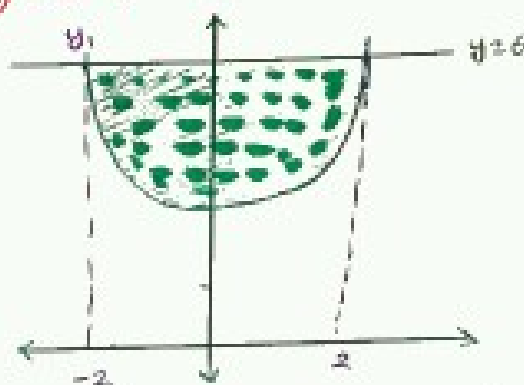
المساحة



$$\text{Area} = \int_a^b [f(x) - g(x)]$$

Ex: find the area of the region (منطقة) in the close

$y_1 = x^2 + 2$  and  $y_2 = 6$ .



نقطة التقاطع بينهما (عدد يتقاسم)

نقطة التقاطع بينهما (عدد يتقاسم)

$$y_1 = y_2 \Rightarrow x^2 + 2 = 6$$

$$x^2 = 4 \Rightarrow x = \pm 2$$

$$\text{Area} = \int_{-2}^2 (6 - x^2 - 2) = \int_{-2}^2 (4 - x^2) \cdot dx = 4(2 - -2) - \frac{x^3}{3} \Big|_{-2}^2$$

$$16 - \frac{1}{3} (x^3 \Big|_{-2}^2) \Rightarrow 16 - \frac{1}{3} (8 - -8) = 16 - \frac{16}{3}$$

$$= \frac{48 - 16}{3} = \frac{32}{3}$$

Q: multiple:- Given the graph below  $f(x)$  find  $\int_0^3 f(x)$

Remark:

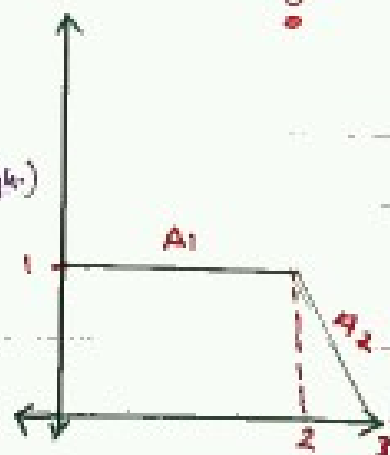
Area of right triangle =  $\frac{1}{2} (\text{base}) * (\text{height})$

" " (مربع) = long \* width

$$A_1 = \frac{1}{2} (2) * (1) = \frac{1}{2}$$

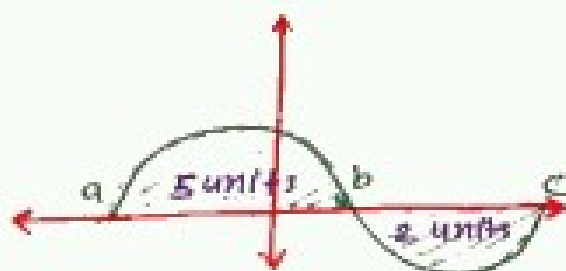
$$A_2 = (1) (2) = 2$$

$$\therefore A = 2 + \frac{1}{2} = \frac{5}{2}$$



# \* Relationship between Area and integrals.

MOHAMMAD  
AL\_ZOUBI



① Find Area of shaded



$$\text{Area} = 5 + 2 = \boxed{7}$$

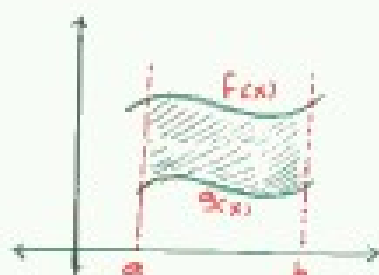
②  $\int_a^c f(x) \cdot dx$

$$\int_a^b f(x) \cdot dx + \int_b^c f(x) \cdot dx$$

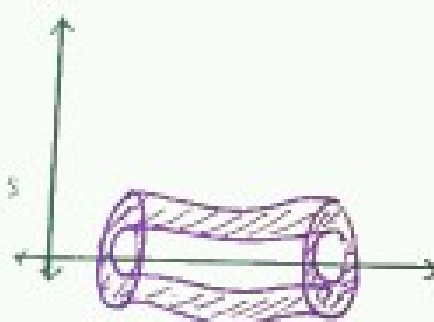
$$5 + -2 = \boxed{3}$$

## VOLUMES

الحجوم

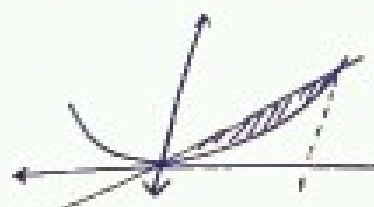


دوران  
Revolving  
about x-axis



$$\text{Volume} = \pi \int_a^b (f(x)^2 - g(x)^2) \cdot dx$$

Ex: Find the volume of the solid (جسم) generated by (تدوير) revolving (دوران) the region bounded by the curve  $y=x$  and  $y_1=x^2$  about the x-axis?



Solution:  $V = \pi \int_a^b (f(x)^2 - g(x)^2) \cdot dx$

Intersection:  $x^2 = x$

$$x^2 - x = 0 \Rightarrow x(x-1) = 0$$

$$x=0, x=1$$

$$\therefore V = \pi \int_0^1 (x)^2 - (x^2)^2 = \pi \left( \frac{x^3}{3} - \frac{x^5}{5} \right)$$

$$V = \frac{2\pi}{15} \text{ units}$$

(H.W): Find the volume of the solid generated by revolving the region that bounded by:  $y = \sqrt{x}$ ,  $\frac{x\text{-axis}}{y=0}$ , and  $x=2$  about the  $x$ -axis.

\*  $x\text{-axis} \Rightarrow y=0$

\*  $y\text{-axis} \Rightarrow x=0$

$$V = \int_0^2 (\sqrt{x})^2 - 0 \, dx = \int_0^2 x \cdot dx = \frac{x^2}{2} \Big|_0^2$$

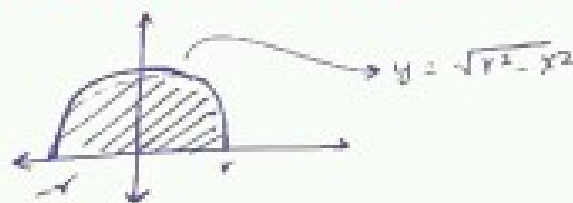
$$V = \frac{1}{2} (4 - 0) = \underline{\underline{2 \text{ unit}}}$$

MOHAMMAD  
AL-ZOUBI

### Application

ملاحظة: دورة نصف محيط كروية

Ex: Find the volume of the sphere of the radius.  
(sphere كروية)



$$V = \pi \int_{-r}^r \left[ (\sqrt{r^2 - x^2})^2 - 0 \right] \cdot dx$$

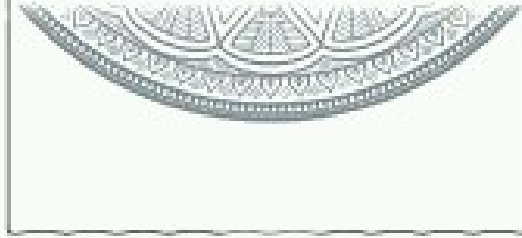
$$= \pi \left( r^2 - \frac{x^2}{2} \right) \Big|_{-r}^r$$

$$= \frac{4}{3} \pi r^3$$

انتشرت مادة الكولاس

... 102A ...

Mohammad Omar Al-Zoubi



# Academic Team OF Healing

Batch  
16



• شفاء

---